Data in Brief 48 (2023) 109218

Contents lists available at ScienceDirect

Data in Brief

journal homepage: www.elsevier.com/locate/dib

Data Article

Dataset of an energy community with prosumer consumption, photovoltaic generation, battery storage, and electric vehicles

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ARTICLE INFO

Article history:
Received 23 February 2023
Revised 17 April 2023
Accepted 4 May 2023
Available online 9 May 2023

Dataset link: A complete energy community dataset with photovoltaic generation, battery energy storage systems and electric vehicles (Reference data)

Keywords:
Battery storage systems
Electric vehicles
Energy community
Photovoltaic energy

ABSTRACT

A challenge that consistently arises when reviewing and justifying novel energy models and theorems is the accuracy of the electrical data used. Therefore, this paper presents a dataset representing a complete European residential community based on real-life data. In this case, a community of 250 residential households was constructed with actual energy consumption and photovoltaic generation profiles collected by smart meters in households in different European locations. Additionally, 200 members of the community were ascribed with their photovoltaic generation, while 150 were owners of a battery storage system. New profiles were generated from the sample collected and were randomly given to each end-user according to their previously defined characteristics. Furthermore, one regular and one premium vehicle were allocated to each household – a total of 500 electric vehicles – with information on their capacity, state of charge, and usage. Moreover, data on the location, type, and prices of public electric vehicle chargers were specified.

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https://doi.org/10.1016/j.dib.2023.109218
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Specifications Table

Subject:	Electrical and Electronic Engineering
Specific subject area:	The data describes an energy community (EC) comprised of residential buildings equipped with photovoltaic (PV) energy generation, battery energy storage system (BESS), and electric vehicles (EV).
Type of data:	Table (.xlsx format)
How the data were acquired:	The data was obtained from a mix of existing publicly accessible datasets plus data obtained by the authors. The public consumption datasets were sourced by meters from households in London, United Kingdom (UK) [1] and Sceaux, France [2], between November 2011 and February 2014, and December 2006 and November 2010, respectively. Moreover, the filtered energy profiles based on these datasets was procured in [3]. The authors also extracted the consumption and photovoltaic generation data from the research group building (GECAD) every ten seconds using the Saia 3-phase energy meters (ale3d5f) linked to a Supervisory control and data acquisition (SCADA) system [4]. In addition, the authors attained data on the Portuguese Contractual Power and Time of Use Tariffs [5,6]. In the case of the battery energy storage system, the authors procured information on the LG home batteries [7] and Energias de Portugal's (EDP's) Premium home Charger [8]. Moreover, data on the different characteristics of several electric vehicles [9] and public chargers [10] was acquired.
Data format:	Raw Filtered
Description of data collection:	The information was gathered from many sources, one for each data type required. This dataset presents energy consumption [1–4] and photovoltaic generation [3] profiles. While also gathering data on battery storage systems [7], electric vehicles [9], electricity contracts and tariffs [5,6], and electric vehicle chargers [10].

Data source location:	**Primary data source**
	• **Renewable energy production profiles**
	  • Institution: GECAD - Research Group on Intelligent Engineering and Computing for Advanced Innovation and Development
	  • City/Town/Region: Porto
	  • Country: Portugal
	• **End-user profiles**
	  • Institution: GECAD - Research Group on Intelligent Engineering and Computing for Advanced Innovation and Development
	  • City/Town/Region: Porto
	  • Country: Portugal
	• **Filtered energy profiles**
	  • Institution: GECAD - Research Group on Intelligent Engineering and Computing for Advanced Innovation and Development
	  • City/Town/Region: Porto
	  • Country: Portugal
	Secondary data used to create the dataset
	• **End-user profiles:**
	  • Institution: UK Power Networks
	  • City/Town/Region: London
	  • Country: United Kingdom
	  • Institution: Électricité de France S.A. - Research, Development and Innovation
	  • City/Town/Region: Sceaux
	  • Country: France

Data accessibility:	**Full dataset described in this publication** • Repository name: Zenodo • Direct URL to data: https://zenodo.org/record/7602546

(*continued on next page*)

Secondary data used to create the dataset

- **Renewable energy production profiles:**
  - Repository name: PES-ISS
  - Direct URL to data: https://site.ieee.org/pes-iss/data-sets/
- **End-user profiles:**
  - Repository name: UK Power Networks
  - Direct URL to data: https://data.london.gov.uk/dataset/smartmeter-energy-use-data-in-london-households
  - Repository name: University of California, Irvine - Machine Learning Repository
  - Direct URL to data: https://archive.ics.uci.edu/ml/datasets/individual/household/electric/power/consumption
  - Repository name: PES-ISS
  - Direct URL to data: https://site.ieee.org/pes-iss/data-sets/
- **Filtered energy profiles:**
  - Repository name: Zenodo
  - Direct URL to data: https://zenodo.org/record/6778401
- **Battery storage systems and electric vehicles:**
  - Repository name: Zenodo
  - Direct URL to data: https://zenodo.org/record/4737293
  - Repository name: Elektrische Voertuigen Database
  - Direct URL to data: https://ev-database.org/nl

Instructions for accessing these data
The data does not have access controls; the readings are open-access and do not compromise the anonymity of the editors and reviewers.

Value of the Data

- With the inclusion of electric vehicles (EVs) and battery energy storage systems (BESS), this dataset is intended for use in sophisticated energy management models for energy communities and smart buildings. Its application in demand response transactive energy models, and optimization algorithms targeted for energy management for smart buildings and data on energy consumption and generation may be relevant in further studies.
- This dataset is helpful for real-life energy community modeling and optimization as it includes data on several essential components, namely renewable energy sources, such as photovoltaic (PV) production and the use of BESSs and EVs. The presence of BESSs and EVs may offer valuable insights into energy usage and assist researchers in developing new technologies and methods for renewable energy integration into the grid. The use of BESSs and EVs can improve grid stability and reliability by enabling the minimization of PV generation uncertainty. In addition, it could be used to generate and test different energy communities with various configurations, such as with or without EVs, and the number of EVs, BESSs, or community members.
- This dataset benefits researchers and anyone interested in the energy sector by providing a valuable data source. Namely, for the application of algorithms and models, primarily through simulation applications, while assisting in the analyzation of the dynamics and structure of a realistic home load profile.
- The European Union (EU) has proposed several policies encouraging transactive energy solutions to combat the hurdles of the rising energy demand and prices and the inclusion of distributed renewable energy sources into the mix. This dataset is essential due to its ability to simulate and manage a typical household every 15 minutes while aiding with the sustainability of the community. In addition, the dataset is helpful for peer-to-peer models, where end-users (250) and EVs (500) could interact, forming a total of 750 players that could participate in peer-to-peer energy transactions. With the simulation of these types of markets, it is possible to reduce the uncertainty provoked by distributed renewable energy sources.

1. Objective

The new advances in smart grids, namely energy communities, open the opportunity to the conception of novel energy management and optimization models, such as to increase community sustainability, optimize the energy balance, and maximize the use of distributed renewable sources. This dataset represents an energy community composed by energy consumers, energy producers, electric vehicles, and distributed energy storage systems. The dataset also considers public charging stations.

2. Data Description

The excel spreadsheet comprises the features of the 250 end-users, namely their energy consumption profiles, characteristics, and allocation of the BESSs and EVs. Generally, the data is directly correlated to each player and can be presented as 15-minute interval data, representing a full day (i.e., 96 periods). Each excel sheet's content is explained below:

- "General Information" sheet: information regarding all 250 players, the BESS, charger, and EV characteristics;
- Players Characteristics: random appointment of the PV production and BESS (0 – not allocated; 1 - allocated) (columns B and C) for each of the 250 players (column A). In addition, the nomination of the contractual power term and respective daily prices (columns E12-25 and F12-25) is presented in kilo-volt-amperes (kVA) and euro (EUR €), respectively. These are distributed to each player (column G12-25). Furthermore, the time of use tariff prices and timetable (for triple and quadruple prices) are defined in EUR (columns E27-37 to H27-37);
- Energy Storage Systems and Charger Characteristics: energy storage systems characteristics in terms of Type, Capacity (kilowatt (kW)), Charge (kW), Discharge (kW), Efficiency (%), Model, and Units randomly distributed (columns K3-13 to Q3-13). The Premium Charger EDP (columns K15-21 to M15-21) presents EDP's four types of premium chargers and their capacity (kW). Moreover, the players were distributed randomly between Type 2 and 3 due to the constraints from the EVs;
- Electric Vehicle Characteristics: EV models features of both regular (columns T3-20 to AB3-20) and premium EVs (columns T22-34 to AB22-34), namely Model, Identification (ID), Brand, Model, Capacity (kW), Charge (kW), Discharge (kW), Efficiency, Consumption (watt-hours per km (Wh/km)) and Units. Also, the EV1 and EV2 departure and arrival periods (15-minute intervals) were presented (columns AD to AK);
- "Mobilidade Elétrica (MOBIe) EV chargers" sheet: publicly available EV chargers in Porto and Vila Nova de Gaia, and their characteristics (columns A to O), namely their ID, User Identifier (UID) of Charger, Type of Charging, State of Charger, City Address, Operator, Capacity (kW), Voltage Level, Price per charge (EUR), Price per minute (EUR/min), Price per kWh per usage (EUR/kWh), Price per hour (EUR/h), Price per kilowatt-hour (kWh) (EUR/kWh);
- "EVs" sheet: regular (EV1) and premium (EV2) EV models distribution between each of the 250 households (rows 1 to 3), and their characteristics (regular: rows 6 to 25; premium: rows 27 to 46). These characteristics are their Model Capacity (kW), charge (kW), Discharge (kW), Efficiency, Initial battery level (kW), Final battery level (kW), Consumption (Wh/km), Departure period, Arrival period, Distance (kilometer (km)), Consumption (kWh), Morning Trip duration (periods), Afternoon Trip duration (periods), Trip duration (periods), Charger Type, Public Charger, Power (kW), and Price at Public Charge Station (EUR);
- "Load" sheet: total energy load by each end-user, in kW, of the 250 residential consumers (columns B to IQ) for one day (96 periods of 15 minutes);

- "PV" sheet: PV energy Production profiles, in kW, of the 250 residential producers (columns B to IQ) for one day (96 periods of 15 minutes);
- "BESS" sheet: Battery energy storage systems model allocation and features (rows 1 to 8), namely their Model Capacity (kW), charge (kW), Discharge (kW), Efficiency, Initial charge (kW), Final charge (kW);
- "Buy Price" sheet: purchasing prices (EDP commercial retailer) for each user (columns B to IQ) at each period for one day (96 periods of 15 minutes);
- "Sell Price" sheet: Feed-in price in Portugal (2023) for each prosumer (columns B to IQ), at each period for one day (96 periods of 15 minutes);
- "Limits" sheet: maximum and minimum CP limits, Power Buy (kW), Power Sell (kW), Fixed costs (EUR), Initial CP (kW), Premium Charger EDP, Sum (kW), New CP level (kW), Fixed costs (EUR);
- "EV1 Buy Price" sheet: purchasing prices for EV1 for each user (columns B to IQ) at each period for one day (96 periods of 15 minutes);
- "EV2 Buy Price" sheet: purchasing prices for EV2 for each user (columns B to IQ) at each period for one day (96 periods of 15 minutes);
- "EV 1 Load" sheet: power used by EV1 for each user (columns B to IQ) at each period for one day (96 periods of 15 minutes);
- "EV2 Load" sheet: power used by EV2 for each user (columns B to IQ) at each period for one day (96 periods of 15 minutes);
- "Max EV1 Dis-Charge" sheet: maximum power usage limit by EV1 for each user (columns B to IQ), at each period for one day (96 periods of 15 minutes);
- "Max EV2 Dis-Charge" sheet: maximum power usage limit by EV1 for each user (columns B to IQ), at each period for one day (96 periods of 15 minutes);
- "EV1 at Home (x)" sheet: location of EV1 (1 – home; 0 – not home) for each user (columns B to IQ), at each period for one day (96 periods of 15 minutes);
- "EV2 at Home (x)" sheet: location of EV2 (1 – home; 0 – not home) for each user (columns B to IQ), at each period for one day (96 periods of 15 minutes);
- "EV1 at Charging Station (x)" sheet: location of EV1 (1 – charging station; 0 – not at charging station) for each user (columns B to IQ), at each period for one day (96 periods of 15 minutes);
- "EV2 at Charging Station (x)" sheet: location of EV2 (1 – charging station; 0 – not at charging station) for each user (columns B to IQ), at each period for one day (96 periods of 15 minutes).

3. Experimental Design, Materials and Methods

The dataset was produced considering the load consumption, PV generation, and BESS in an idealized typical energy community household. The household profiles were created from a sample of energy profiles undergoing a normalization process and consequently allotted a random maximum consumption and generation level. In addition, the profiles were multiplied to obtain the total number of profiles needed to complete the community (250 end-users). Likewise, the household characteristics were uniformly distributed within the defined range.

Of the 250 total end-users, 200 members of the community were credited with PV generation profiles, and 150 were allocated a BESS. Moreover, each household was given two EVs (regular and premium models). The process of ascribing these profiles was done randomly and uniformly.

3.1. Load Consumption

The consumption profiles were selected from European household profiles to represent a typical energy community. These sample profiles were normalized and given a contracted power

level. A random and uniform dispersal of end-users between the available contractual power levels was applied. Due to the high energy usage by households, EVs, and BESS, only the contracted powers [13.8, 17.25, 20.7, 27.6] kW were chosen. Also, these were multiplied by 95%, as consumers usually do not hit their maximum contracted power values. Furthermore, at each period of 15 minutes, the household consumption was multiplied by a random uniform value between [0.8, 1.1] to help in the variability of the profiles created. Resulting in the total energy load by each end-user, in kW, of the 250 residential consumers for one day (i.e., 96 periods of 15 minutes).

3.2. Photovoltaic Production

A sample generation profile was collected to depict the 200 community members as all the households in this hypothetical situation reside in the same general location. In addition, this profile was normalized, multiplied by a maximum generation level, and uniformly dispersed. Moreover, each period was multiplied by a random uniform number between [0.8, 1.1] to aid with the variation of the profiles produced.

3.3. Electric Vehicles

A survey of EVs was made, considering their features and popularity. These were divided between regular and premium EVs, reflecting their average market price. The EV data is calculated directly from each vehicle's trip distance, where it determined the EV consumption per distance covered, according to Tables 1 and 2. Finally, each household was given a regular (EV1) and a premium (EV2) EV to spread randomly and uniformly among the end users. The amount and features of regular and premium EVs are displayed in Tables 1 and 2.

Table 1 exhibits the 250 regular EVs selected compared to the 250 premium models in Table 2 due to their lower characteristic values. The vehicles in Table 1 with the highest battery capacity are the VW ID3 pro and the ID3 pro performance with 55.7 kW, lower than the lowest battery capacity in Table 2, the Audi e-tron with 63 kW. The same can be said about the Ford AWD in Table 2, with the lowest charge/discharge capacity of 90.5 kW, compared to the highest in Table 1, the same VW ID3 pro and the ID3 pro performance with 75.2 kW. Additionally, the consumption of these vehicles is similar, being between [119–224] Wh/km, averaging around 161 Wh/km.

Table 1
Number of regular EV units and their features.

	Brand	Capacity (kW)	Charge / Discharge (kW)	Efficiency (%)	Consumption (Wh/km)	Units
Regular	Dacia Spring	27.4	27.7	0.85	119	22
	Fiat 500e	36.2	64.6	0.96	120	18
	Opel Corsa-e	40.9	74	0.86	136	15
	MG Range	41.8	48	0.94	186	16
	VW ID3 pure	43.5	69.3	0.95	132	14
	Renault Zoe	44	40.6	0.81	177	14
	Mazda MX30	30	33.4	0.97	160	17
	Citroën e-C4	40.9	74	0.86	129	18
	VW ID3 pro	55.7	75.2	0.92	140	15
	Opel Mokka	42.9	74	0.9	140	19
	Peugeot e-GT	44	55	0.97	151	17
	Nissan Leaf	36	36.7	0.93	137	19
	Honda e	26.8	32.6	0.96	180	16
	Kia e-Soul 64	39.2	35.7	0.92	157	17
	VW perf.	55.7	75.2	0.92	142	13

Table 2
Number of premium EV units and their features.

	Brand	Capacity (kW)	Charge / Discharge (kW)	Efficiency (%)	Consumption (Wh/km)	Units
Premium	BMW i4	80.7	121	0.94	137	31
	Audi Quattro1	76.6	92.9	0.98	186	26
	Mercedes 4M1	66.5	101.6	0.95	168	24
	Audi Sport	63.6	111.7	0.93	216	19
	Audi e-tron	63	111.7	0.92	224	31
	Audi Quattro2	76.6	92.9	0.99	182	24
	Mercedes 4M2	64.8	101.6	0.93	159	27
	Ford AWD	85.6	90.5	0.92	164	18
	BMW iX3	74.3	111.8	0.91	161	21
	Mercedes EQ	80	94.6	0.92	223	29

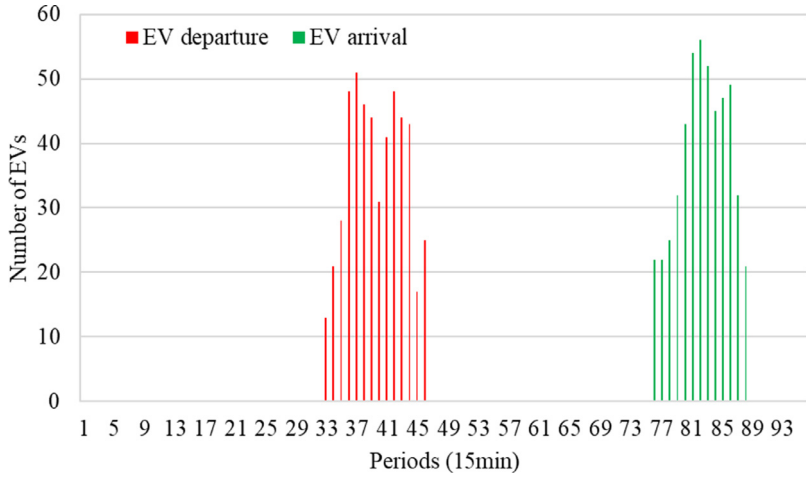

Fig. 1. Number of EVs leaving and returning to the household.

In Tables 1 and 2, the efficiency refers to the EV's battery charging/discharging process. The efficiency is calculated by the ratio between the energy needed to charge the EVs battery and the energy effectively stored in the vehicle. This results from the EV battery's charging/discharging process losses and is generally above 90%.The EV's departure and arrival times were dispersed uniformly to each EV and household. The available departure and arrival times were restricted within the intervals [08:15–11:30] and [19:00–22:00], respectively. Fig. 1 depicts the number of EVs leaving and returning to the household.

Knowing the usual patterns of a typical household, most consumers utilize their EVs as a means of transportation to and from work or school. Fig. 1 portrays the number of EVs in each period of 15 minutes, randomly leaving and returning to the household. Visualizing the present graph, the departure peak is at 09:15, with 51 vehicles leaving the household, while 56 vehicles return home at 20:30.The premium home charger from EDP, specifically the 7.4 kW and 11 kW models, was chosen due to the average EV consumption demand. The models were uniformly and randomly allocated to the whole community. In addition, the energy consumed readings by the vehicles at home were individually obtained and stored independently from the end-user's household consumption profile. Fig. 2 depicts the number of EVs parked at home and the charging station.

As seen in Fig. 2, we verify the usual patterns of a typical day, with the when and which EVs are parked at home or the charging station. All 500 EVs in the community are parked at home during the nighttime periods. However, there is an increase in EVs parked in charging

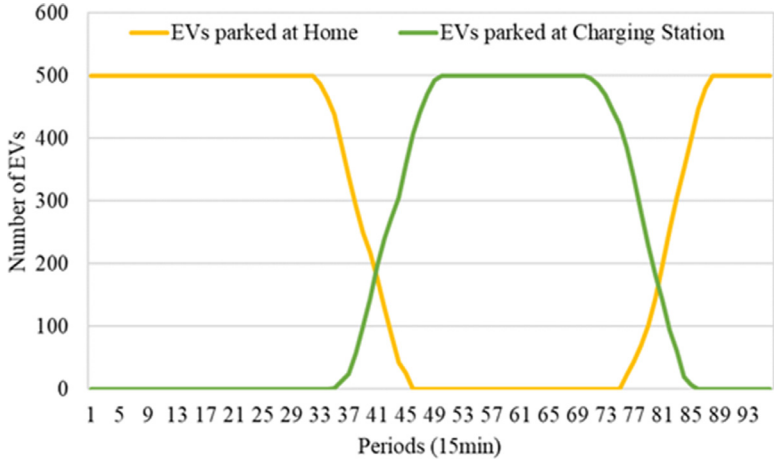

Fig. 2. Number of EVs parked at home and the charging station.

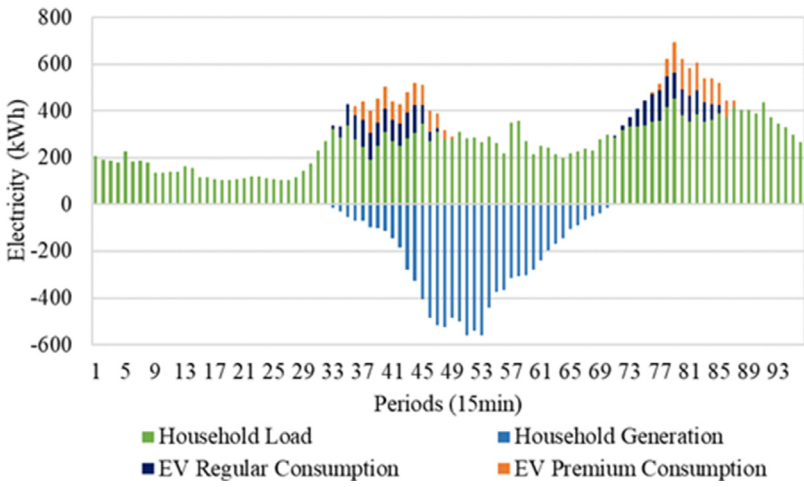

Fig. 3. Households' load, households' generation, and premium/regular EV consumption.

stations, while EVs parked at home have decreased. This is due to the morning usage of EVs as transportation to work or school. Inversely, with EVs returning home, there is a reduction in EVs parked at charging stations and an increase at home. To better visualize the aggregation of the dataset elements, the sum of the households' load, the households' generation, and the premium/regular EV consumption are depicted in Fig. 3.

As can be seen, Fig. 3 depicts the total amount of energy in kWh used over 24 hours, broken down into 15-minute periods, individually illustrating the load, the premium/regular EV load, and PV production. We can identify in green the energy consumption patterns, namely a noticeable increase in morning usage as consumers begin using appliances, followed by a decrease throughout the day due to work or school, then another increase in evening usage as consumers arrive home and use their appliances. In blue, we verify a spike in PV generation during the higher solar exposure hours. In dark blue and orange, the EV load consumption is visible due to the EVs being used leaving and arriving home.

An extensive search was performed on the accessible public EV chargers in Porto and Vila Nova de Gaia [10]. Furthermore, the principal characteristics of these public chargers were listed.

3.4. Energy Storage Systems

The energy storage systems selected to be allocated to the 150 end-users were obtained from the LG catalog, namely variations of the Residential Energy Storage Unit (RESU) model [7]. The models and respective characteristics were randomly and uniformly scattered between the households.

3.5. Pricing Tariffs

The values used for the energy pricing for end customers are the referenced standard low voltage tariffs applied in Portugal [5,6]. In this case, the prices are fixed and obtained from the practiced tariffs by the Serviço Universal (SU) retailer for small-scale electricity users. In addition, the tri-hourly tariff was chosen due to having a 45% reduction in the price per kWh expended during off-peak hours. While the PV generation sell price is given by the Feed-In Tariff (FiT) in Portugal (45 euro per megawatt-hour (EUR/MWh)) [11].

Ethics Statements

This dataset does not require ethical approval, as the data on the consumption and PV generation of households was obtained through open repositories. In addition, the individual labels and descriptions of the data were removed beforehand by the authors. The data were also normalized to prevent the identification of the specific buildings, and their precise location was not disclosed. The data collection did not include direct human, animal, or social media data, only energy consumption and generation values.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

A complete energy community dataset with photovoltaic generation, battery energy storage systems and electric vehicles (Reference data) (Zenodo)

CRediT Author Statement

Ricardo Faia: Conceptualization, Methodology, Software, Data curation, Writing – original draft, Visualization, Investigation, Validation; **Calvin Goncalves:** Methodology, Data curation, Writing – original draft, Writing – review & editing, Validation; **Luis Gomes:** Supervision, Methodology, Writing – review & editing, Validation; **Zita Vale:** Supervision, Methodology, Writing – review & editing, Validation.

Acknowledgments

This article is a result of the project REal-Time support Infrastructure and Energy management for Intelligent carbon-Neutral smArt cities (RETINA) (NORTE-01-0145-FEDER-000062), supported by Norte Portugal Regional Operational Programme (NORTE 2020), under the PORTUGAL 2020 Partnership Agreement, through the European Regional Development Fund (ERDF). The authors acknowledge the work facilities and equipment provided by GECAD research center (UIDB/00760/2020) to the project team.

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